

Decision Tree Pruning – Complete Notes (Pre-Pruning & Post-Pruning)

1. Why Pruning is Needed

- Decision Trees can grow very deep and memorize training data.
- Deep trees have low bias but high variance (overfitting).
- Training error becomes very low but test error increases.
- Pruning controls model complexity and improves generalization.

2. Overfitting in Trees

- Tree keeps splitting until leaves are pure.
- Creates very small regions.
- Fits noise instead of real pattern.
- Leads to unstable predictions.

3. Pre-Pruning (Early Stopping)

Definition:

- Stop tree growth during training.

Common Parameters:

- max_depth
- min_samples_split
- min_samples_leaf
- max_leaf_nodes
- minimum impurity decrease

Advantages:

- Faster training
- Prevents extremely deep trees

Disadvantages:

- May stop too early
- Risk of underfitting

4. Post-Pruning (Cost-Complexity Pruning)

Definition:

- First grow the full tree.
- Then remove weak branches.

Cost-Complexity Formula:

$\text{Cost} = \text{Error} + \alpha \times (\text{Number of Leaves})$

Where:

- Error = MSE (Regression) or Gini/Entropy (Classification)
- α = Complexity penalty parameter
- Larger $\alpha \rightarrow$ Smaller tree

Decision Rule:

- If pruning reduces total cost, remove subtree.
- Choose optimal α using validation or cross-validation.

5. Numerical Example Summary (Regression)

Example Dataset (Experience vs Salary):

1 $\rightarrow$ 30
2 $\rightarrow$ 35
3 $\rightarrow$ 40
4 $\rightarrow$ 80
5 $\rightarrow$ 85
6 $\rightarrow$ 90

Root Split: Experience ≤ 3.5

Left Mean = 35

Right Mean = 85

Deeper splits reduce MSE further.

If α is small $\rightarrow$ keep deep splits.

If α is large $\rightarrow$ prune deeper nodes.

6. Pre-Pruning vs Post-Pruning Comparison

Pre-Pruning:

- Stops tree early
- Simpler and faster
- Risk: Underfitting

Post-Pruning:

- Grow fully, then prune
- Uses α penalty
- Better complexity control
- Implemented in sklearn using `ccp_alpha`

7. Important Interview Insights

- Pruning reduces variance and slightly increases bias.
- Small $\alpha \rightarrow$ complex tree.
- Large $\alpha \rightarrow$ simple tree.
- Random Forest usually grows deep trees.
- Gradient Boosting uses shallow trees by design.
- Pruning is more critical when dataset is small.